Cell Structure

4.1 Studying Cells

A cell is the smallest unit of a living thing, and every organism, whether it consists of a single cell, like a bacterium, or of trillions of cells, like a human being, is constructed from these fundamental units, which carry out all of the chemical reactions that are necessary for life.

Because most cells are far too small to be seen with the unaided eye, the study of cells depended on the development of the microscope; in 1665 Robert Hooke examined thin slices of cork and described the tiny compartments he observed as cells, a term that biologists continue to use today.

Light microscopes, which pass visible light through a specimen and bend it through a system of lenses, can magnify objects up to approximately 1,000 times, whereas electron microscopes, which use a beam of electrons rather than light, achieve magnifications of up to 1,000,000 times and therefore reveal structures inside the cell that cannot otherwise be resolved.

The cell theory, developed during the nineteenth century by Matthias Schleiden, Theodor Schwann and Rudolf Virchow, states that all living organisms are composed of one or more cells, that the cell is the basic unit of life, and that new cells arise only from the division of pre-existing cells.

4.2 Prokaryotic Cells

Prokaryotes, which include bacteria and archaea, are single-celled organisms that lack a membrane-bound nucleus and other membrane-bound organelles; instead, their genetic material is located in a central region of the cell called the nucleoid.

All prokaryotic cells share four common components: a plasma membrane that separates the interior of the cell from its environment, cytoplasm in which the cellular reactions occur, DNA that carries the genetic instructions, and ribosomes that synthesise proteins.

Most prokaryotes range from 0.1 to 5.0 micrometres in diameter, and their small size is significant because it allows ions and organic molecules that enter the cell to diffuse rapidly to all of its regions, while waste products produced within the cell can likewise diffuse out quickly.

As a cell increases in size, its volume grows faster than its surface area; for a cube with side length L, the ratio is expressed as SA:V = 6/L, which means that a larger cell has relatively less membrane through which to exchange materials with its surroundings.

4.3 Eukaryotic Cells

Eukaryotic cells, which make up animals, plants, fungi and protists, are typically 10 to 100 micrometres in diameter and are characterised by a membrane-bound nucleus together with numerous membrane-bound organelles that allow different metabolic processes to be compartmentalised within the cell.

The plasma membrane, which is composed predominantly of a phospholipid bilayer interspersed with integral and peripheral proteins, cholesterol and glycolipids, functions as a selectively permeable barrier that regulates the transport of ions, nutrients and metabolic waste products, thereby maintaining the homeostatic conditions that are indispensable for cellular viability.

The cytoplasm comprises the entire region of the cell between the plasma membrane and the nuclear envelope, and it consists of organelles suspended in a gel-like cytosol, which is approximately 70 to 80 percent water and contains dissolved glucose, salts, fatty acids, amino acids and proteins.

The nucleus, typically the most prominent organelle in a eukaryotic cell, houses the cell's DNA and directs the synthesis of ribosomes and proteins; it is surrounded by a double membrane called the nuclear envelope, which is perforated by nuclear pores that regulate the passage of molecules between the nucleus and the cytoplasm.

Ribosomes, which are the cellular structures responsible for protein synthesis, are composed of ribosomal RNA and protein, measure approximately 20 to 30 nanometres in

diameter, and are found either free in the cytosol or attached to the cytoplasmic side of the endoplasmic reticulum.

Mitochondria are double-membraned organelles, often described as the powerhouses of the cell, that carry out cellular respiration and thereby produce adenosine triphosphate (ATP), the principal energy-carrying molecule of the cell; notably, mitochondria possess their own DNA and ribosomes.

The endoplasmic reticulum is a series of interconnected membranous sacs and tubules that collectively modify proteins and synthesise lipids; the rough endoplasmic reticulum is studded with ribosomes, whereas the smooth endoplasmic reticulum, which lacks ribosomes, synthesises carbohydrates, lipids and steroid hormones and detoxifies certain medications.

The Golgi apparatus, which consists of a series of flattened membranous sacs, receives vesicles from the endoplasmic reticulum and subsequently sorts, tags, packages and distributes lipids and proteins to their final destinations both inside and outside the cell.

Lysosomes, which are found in animal cells, are membrane-bound organelles containing digestive enzymes that function optimally at an acidic pH of approximately 4.5, and they break down worn-out organelles, macromolecules and engulfed pathogens.

4.4 Features of Plant Cells

In addition to the organelles common to all eukaryotic cells, plant cells possess a rigid cell wall composed principally of cellulose, a polysaccharide made of glucose units, which provides structural support, protects the cell and gives it its characteristic shape.

Chloroplasts are plant cell organelles that carry out photosynthesis, a series of reactions that uses carbon dioxide, water and light energy to produce glucose and oxygen; like mitochondria, chloroplasts contain their own DNA and ribosomes, which supports the endosymbiotic theory of their origin.

The central vacuole, which can occupy as much as 90 percent of the volume of a mature plant cell, stores water and dissolved substances and plays a key role in regulating the concentration of water in the cell, so that when the vacuole loses water the plant wilts.